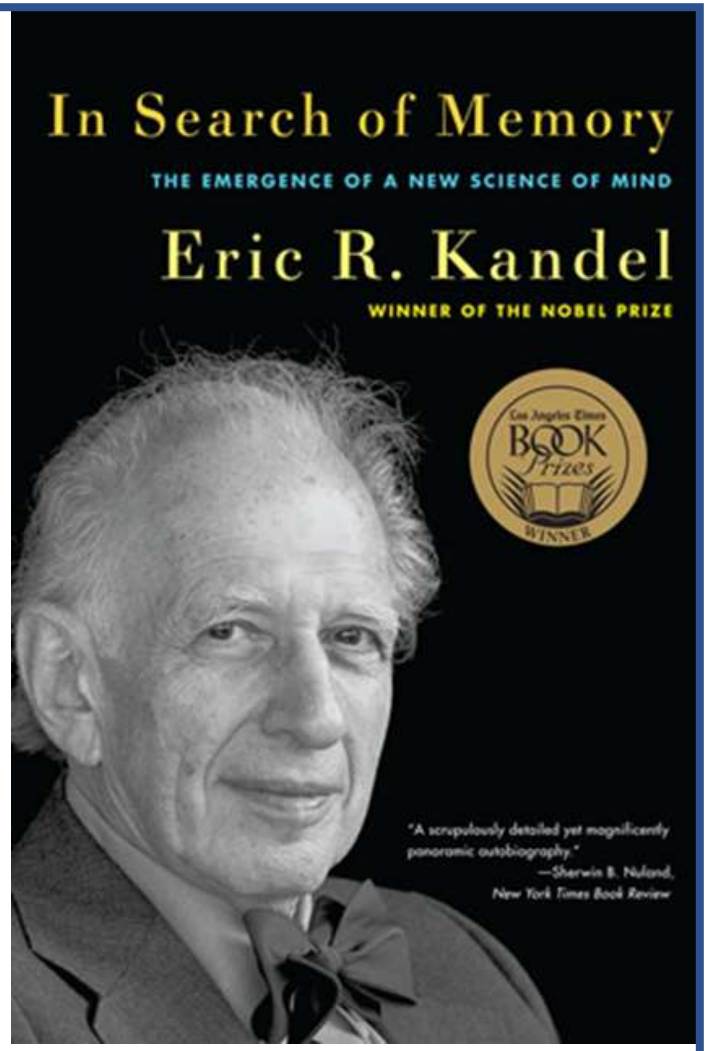
Lecture 8

GFN 1000
In Dialogue with Nature

Eric Kandel

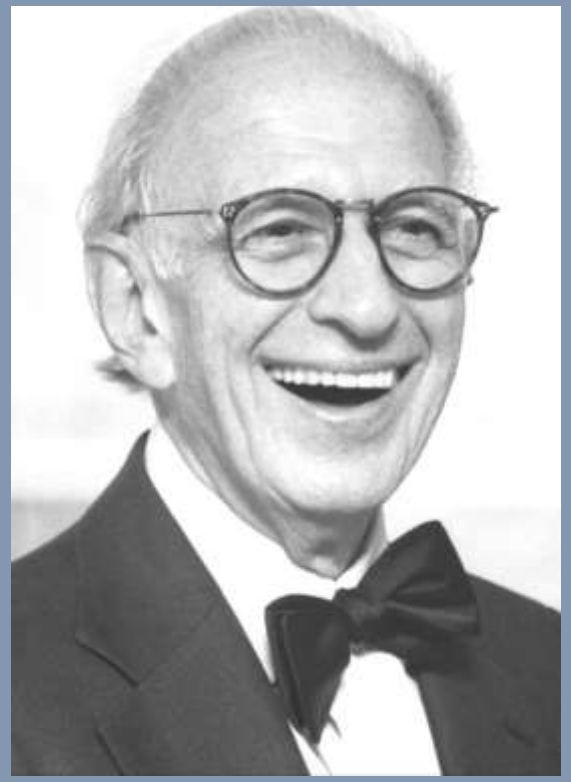
In Search of Memory
The Emergence of a New Science of Mind



- Selected chapters
  - Chapter 4: “One cell at a time”
  - Chapter 28: “Consciousness”
- Partly autobiography
- Partly the intellectual history of the emergence of a new science of mind
  - Understanding the human mind in biological terms at the levels of cells and molecules.

Core Question: What is the human mind?	
Where does the mind come from?	Can it be studied with science?

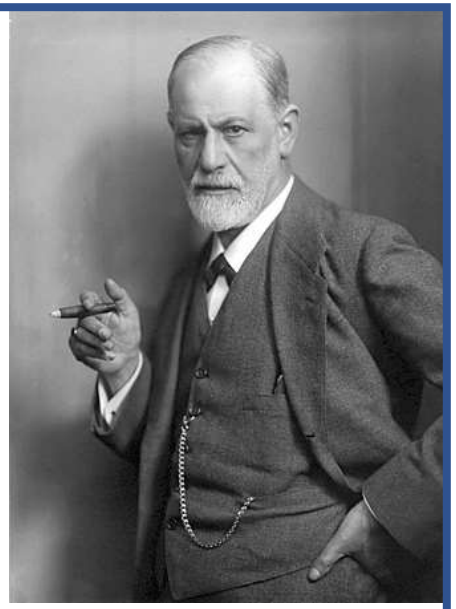
Eric Kandel



1929-

- Born in 1929 in Vienna, Austria.
- Fled to the U.S. in 1939, one year after Hitler came to power.
- Majored (undergraduate) in modern European history and literature at Harvard University.
- Fascinated by psychoanalysis, inspired to study consciousness by Sigmund Freud.

Sigmund Freud



1856 – 1939

- An Austrian neurologist
- MD at the University of Vienna, 1881.
- Founder of psychoanalysis (心理分析).
- Freud lived and worked in Vienna, having set up his clinical practice there in 1886.
- In 1938, Freud left Austria to escape Nazi persecution.
- Died in exile in the United Kingdom in 1939.

Structural theory of mind

- Proposed by Sigmund Freud
- Mental functions: conscious and unconscious (can be known by psychoanalysis)
- Psychic agencies:

  Ego (自我):

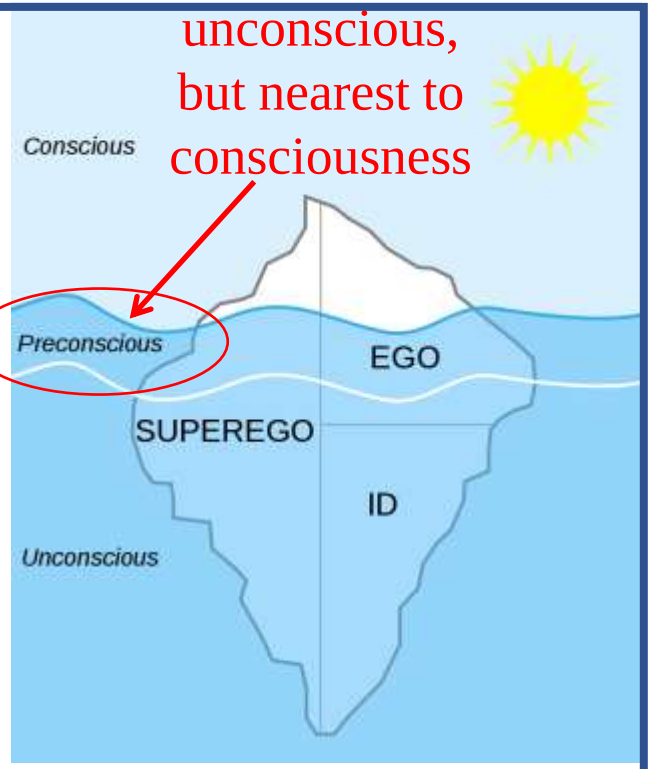
  The “autobiographical self (自述者)”；
  Conscious(connected to the world through the senses)
  Unconscious(repression, denial, sublimation)

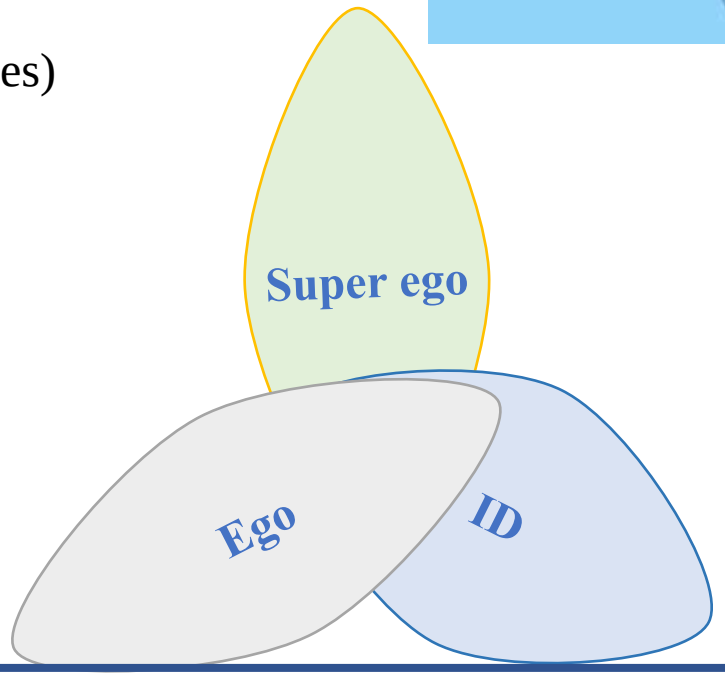
  Id (本我):

  Hedonistic principle (seek pleasure/avoid pain)
  Unconscious

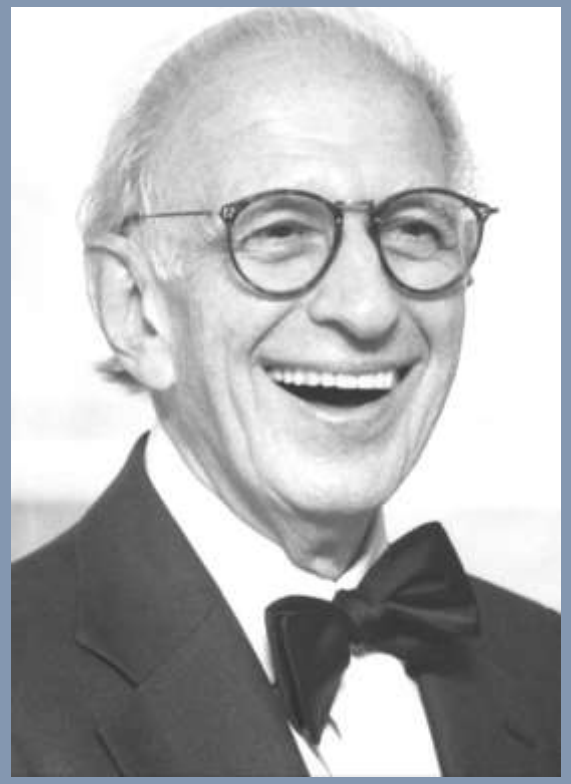
  Superego (超我):

  Moral agency (aspirations)
  Largely unconscious





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Journey in Reverse

- Entered Grundfest's lab at Columbia University in 1955.
- Grundfest's advice: "One cell at a time" (Title of Ch. 4)
- Reductionism (Ch. 4, para. 7)
  - Investigate a complex system (e.g. biological entities) by studying lower-level activities and properties (e.g. physical-chemical entities like atoms and molecules).
  - The idea of reductionism is very influential in the natural sciences.

Journey in Reverse

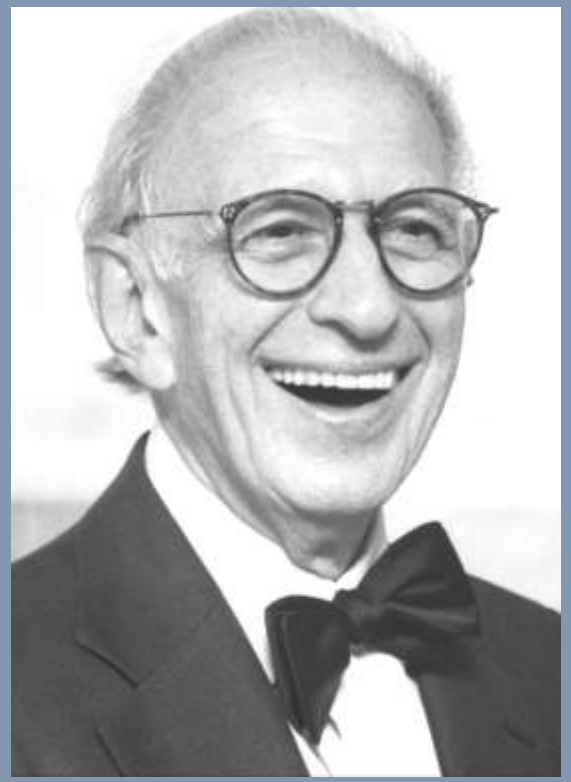
Kandel's research: a reductionistic approach to understand the mechanism of memory at the molecular level

Structural theory of mind
• Top-down approach



Signaling elements of the nervous system
• Bottom-up approach

Eric Kandel

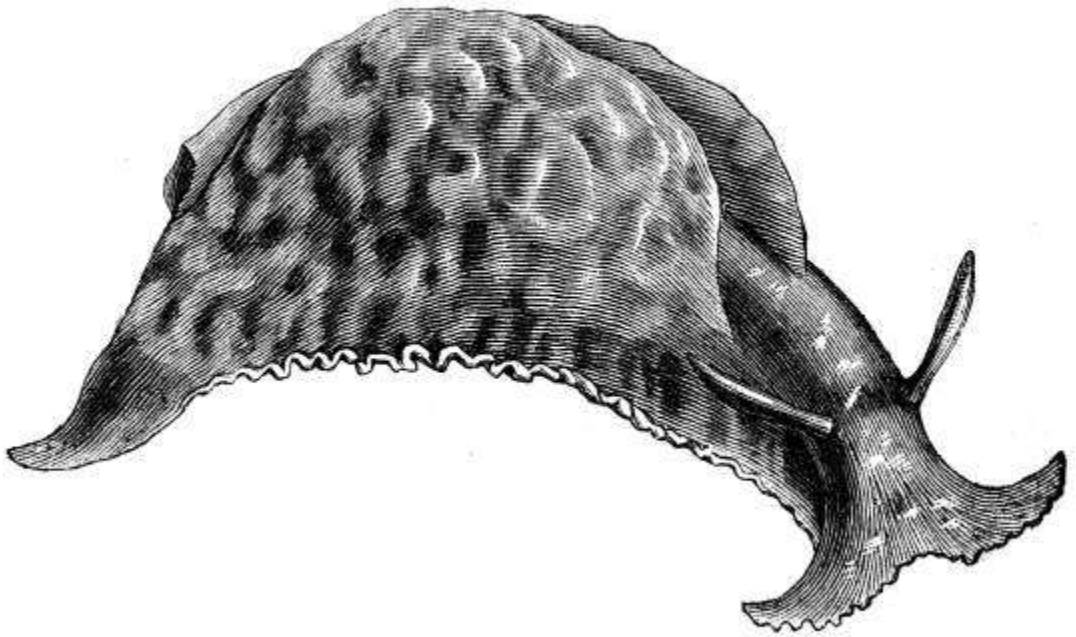


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- Entered Harry Grundfest's lab at Columbia University for a 6-month selective period in 1955.
- Got MD in medical school at New York University in 1956.
- Nobel Laureate in Physiology or Medicine in 2000 for his contribution to the physiological basis of memory storage in neurons.

Aplysia (海兔, 一种螺类)

- One of the **simplest** animals known to be capable of **learning**
- **Large nerve cells** amenable to experimental manipulation
- Uncovered important molecular mechanisms that drive the processes of learning and memory storage.



This Lecture

1. The mind-body problem.
2. The biggest problem in science?
    - Easy problem: unity of consciousness
    - Hard problem: subjectivity
3. Free will?

1. The Mind-body Problem

The Mind-body Problem

The mind-body problem :

- What is the relationship between **mind** and **body**?
- Is there a real division between **mind** and **body**?

Dualism (二元论):

Mind and **body** are two different entities.
(Two kinds of stuff)

Monism (一元论):

Mind and body are the same entity.

Dualism (二元论)

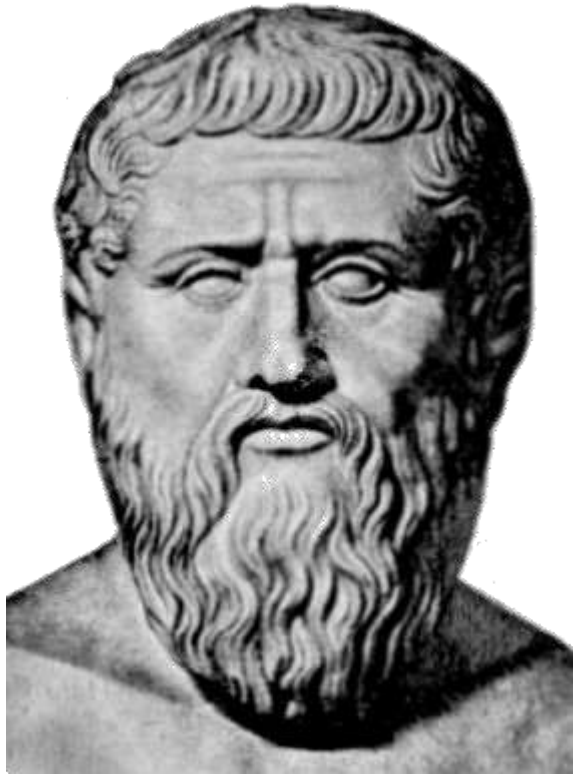
- Plato argued that

Body: from the world of materials, mortal

Soul: from the world of forms, immortal

> Text 1, Para. 1b.34
>
> We are **souls** imprisoned in **bodies**.

The soul and body will separate at death, the former goes back to the world of forms.



Plato

Dualism (二元论)

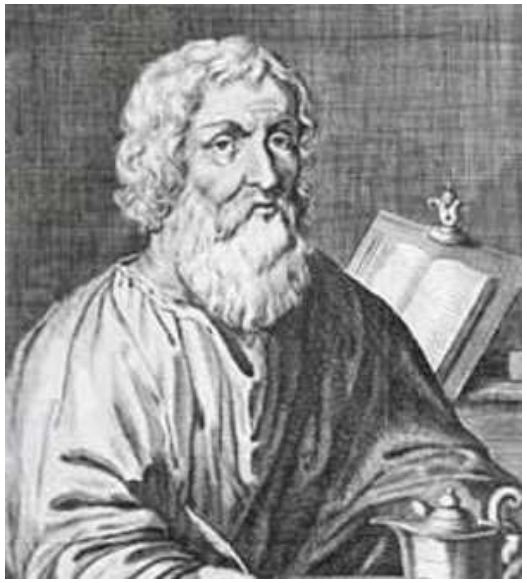
13th Century	Thomas Aquinas (阿奎那): incorporated into Christian thought
17th Century	René Descartes (笛卡尔): body (material substance) + mind (spiritual nature of soul)
20th Century	Karl Popper (波普尔), John Eccles…

Problems asked in Dualism (二元论)

- If mind and body are two different entities, how could mind and body possibly interact? 心灵和肉体怎样地相互作用呢？
  - Direct Intervention of God (Descartes)
  - Pre-established harmony (Leibniz)
- Is it possible to study a nonmaterial mind (scientifically)?

  (科学地)研究一个非物质的心灵，可能吗？

Monism (一元论)



Hippocrates (460-370BC)

Proposed Monism:

> Para. 28.5
> …arguing that all mental processes **emanate from the brain**.

- There is just one sort of substance, **the physical, i.e. matter**.
- Mental events can be completely explained in the terms of physical ones, usually events in the brain.

- ‘Father of Medicine’
- Text 5: Pangenesis (泛生说): heredity = transfer of miniature body parts, embryos were assembled from a set of minuscule components (1, par. 8)

Monism (一元论)

Para. 28.8

Today, most philosophers of mind agree that what **we call consciousness derives from the physical brain**.

- Suggests a **possibility** for the *scientific* study of the mind.
- But this does **NOT** guarantee we can understand consciousness scientifically!

2. The Biggest Problem in Science?

The Biggest Problem in Science?

- Chapter 28: “Consciousness”

“Understanding consciousness is by far the most challenging task confronting science.” (Para. 3, Chap. 28)

Eric Kandel in *In Search of Memory*, 2006

“If you think you have a solution to the problem of consciousness, you haven’t understood the problem.”

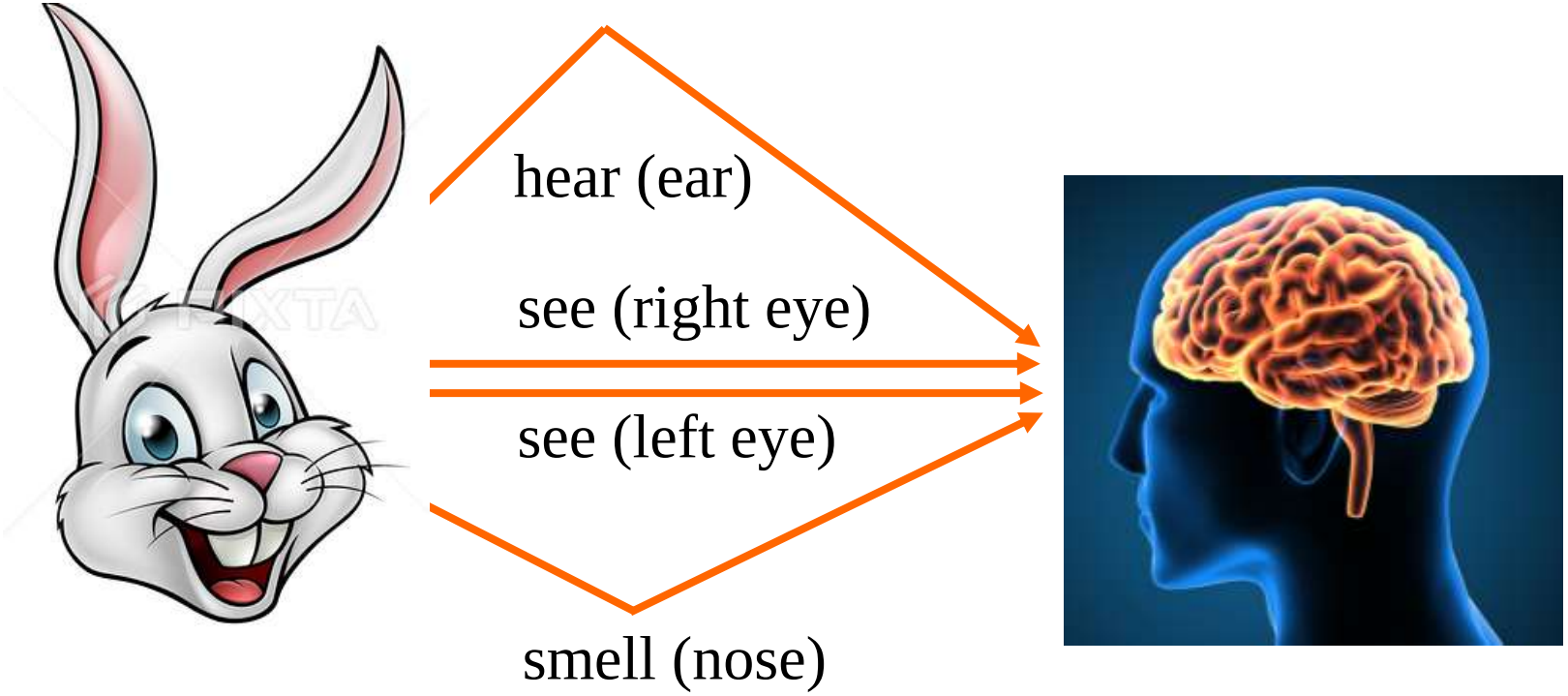
Susan Blackmore in *Consciousness: An Introduction*, 2011

The Biggest Problem in Science?

- Why is it so challenging?
- Two characteristics of consciousness
    a. Easy problem: Unity
    b. Hard problem: Subjectivity

Easy Problem: Unity of Consciousness

The unitary nature of consciousness refers to the fact that our experiences come to us as a unified whole.



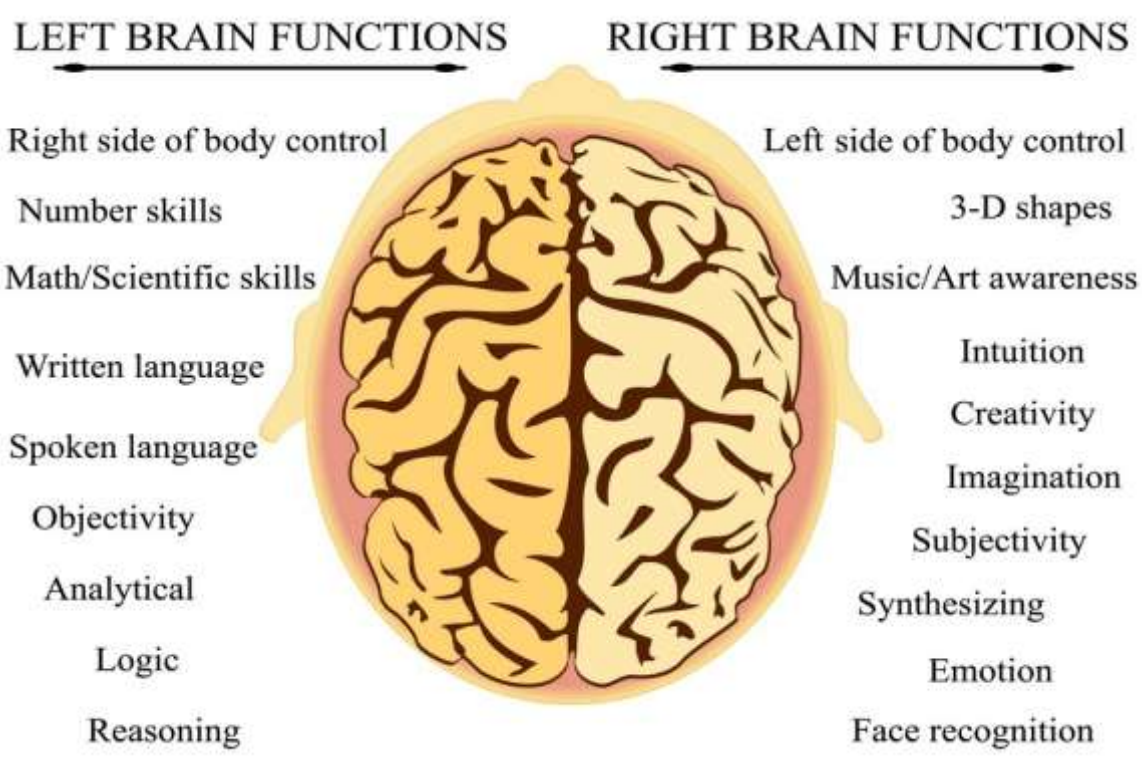
- Signals from 4 different channels.
- There are no 4 consciousnesses.

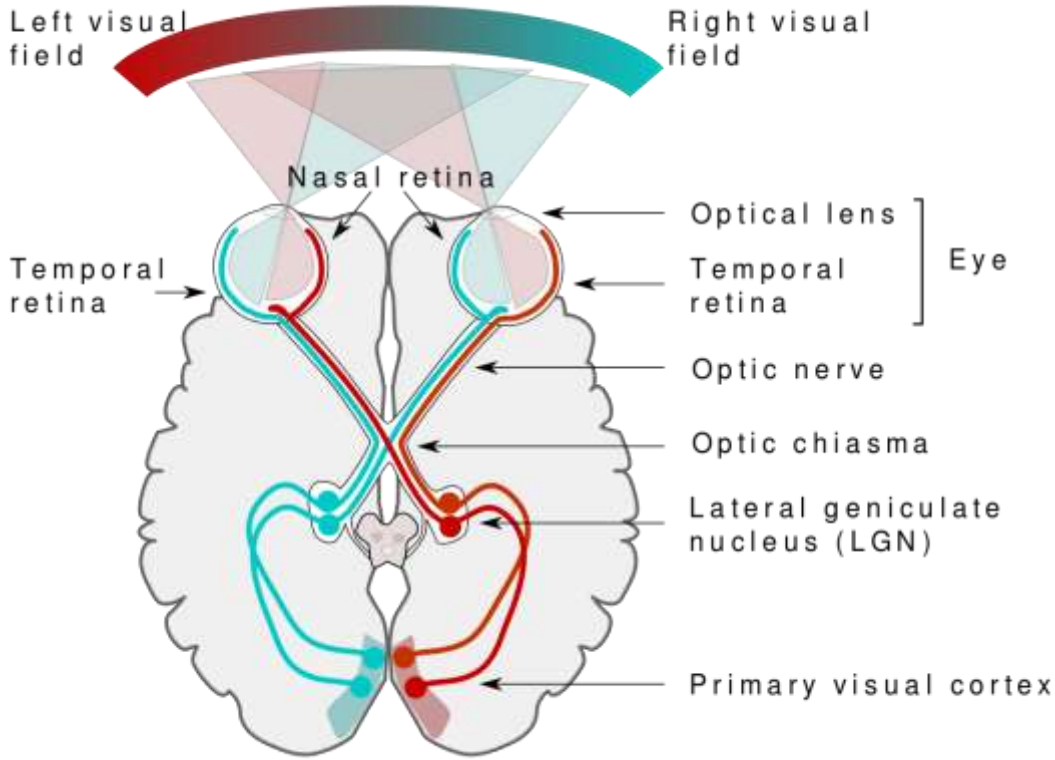
Two Conscious Minds?

- “This unitary nature can break down. In a surgical patient whose brain is severed (切断) between the two hemispheres (半球), there are two conscious minds, each with its own unified percept.” (Chap. 28, Para. 11)
- How?

Split Brain (From BBC's *Brain Story*)




LEFT BRAIN FUNCTIONS
RIGHT BRAIN FUNCTIONS
Right side of body control
Number skills
Math/Scientific skills
Written language
Spoken language
Objectivity
Analytical
Logic
Reasoning
Left side of body control
3-D shapes
Music/Art awareness
Intuition
Creativity
Imagination
Subjectivity
Synthesizing
Emotion
Face recognition


Left visual field
Right visual field
Nasal retina
Temporal retina
Optical lens
Temporal retina
Eye
Optic nerve
Optic chiasma
Lateral geniculate nucleus (LGN)
Primary visual cortex

Lobotomy (脑白质切除术)

- A form of psychosurgery, a neurosurgical treatment of a mental disorder that involves severing connections in the brain's prefrontal cortex (额叶皮层).
- The procedure is to cut the connections to and from, or simply destroy, the prefrontal cortex.
- It often results in major personality changes.
- First performed by Dr. Moniz (winner of the Nobel Prize in Physiology or Medicine, 1949)



Rosemary Kennedy

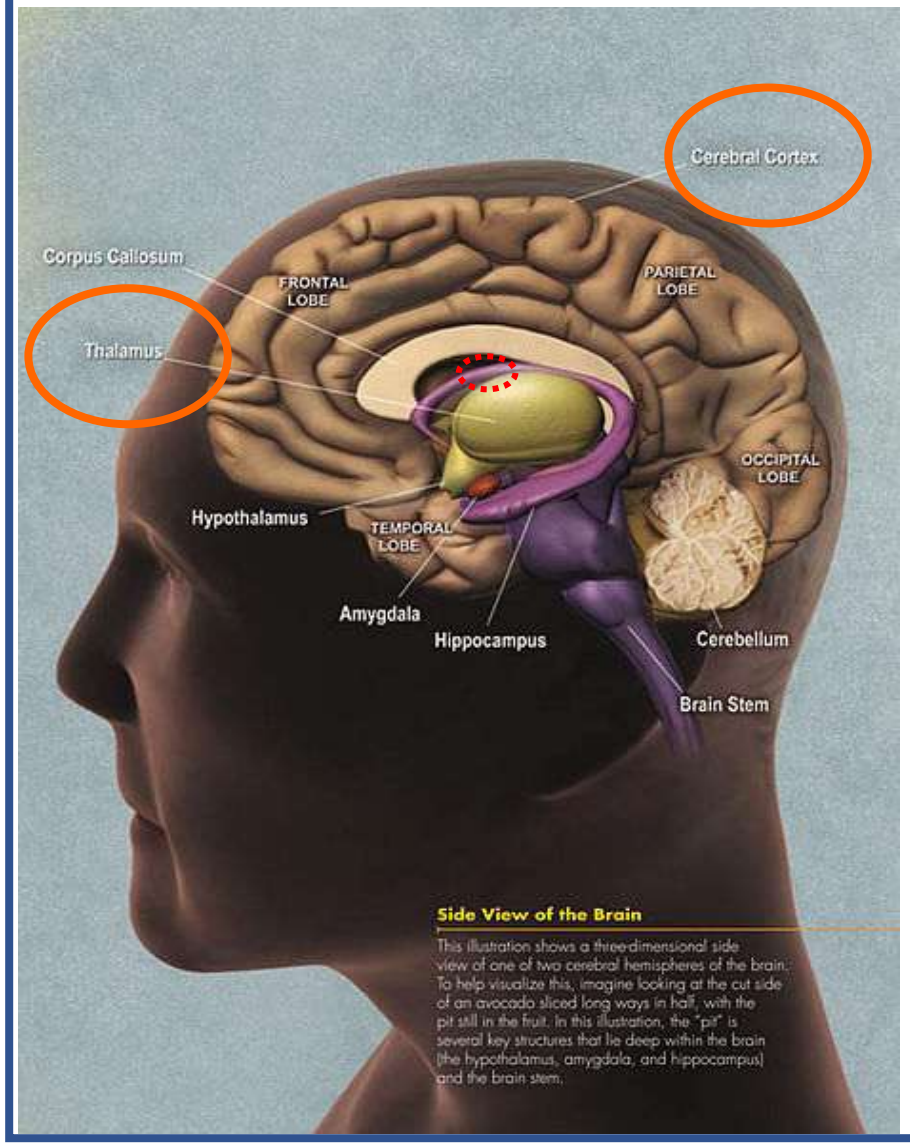


1938-2005

- Sister of President John F. Kennedy
- In her early young adult years, Rosemary Kennedy experienced seizures(癫痫) and violent mood swings.
- In response to these issues, her father arranged a prefrontal lobotomy for her in 1941 when she was 23 years of age.
- The procedure left her permanently incapacitated (无法行动) and rendered her unable to speak intelligibly for the rest of her life.
- According to a survey conducted around 1950, about 1/3 of patients had no change after the surgery, and 1/3 of patients became even worse.

Where does the **unity** of consciousness take place in the brain?

---Two Theories



- Edelman's theory: Widely distributed throughout the cortex (脑皮层) and thalamus (丘脑).
- Crick and Koch's theory: A specific set of neurons. Claustrum (屏状核)? (Chap. 28, Para. 23)

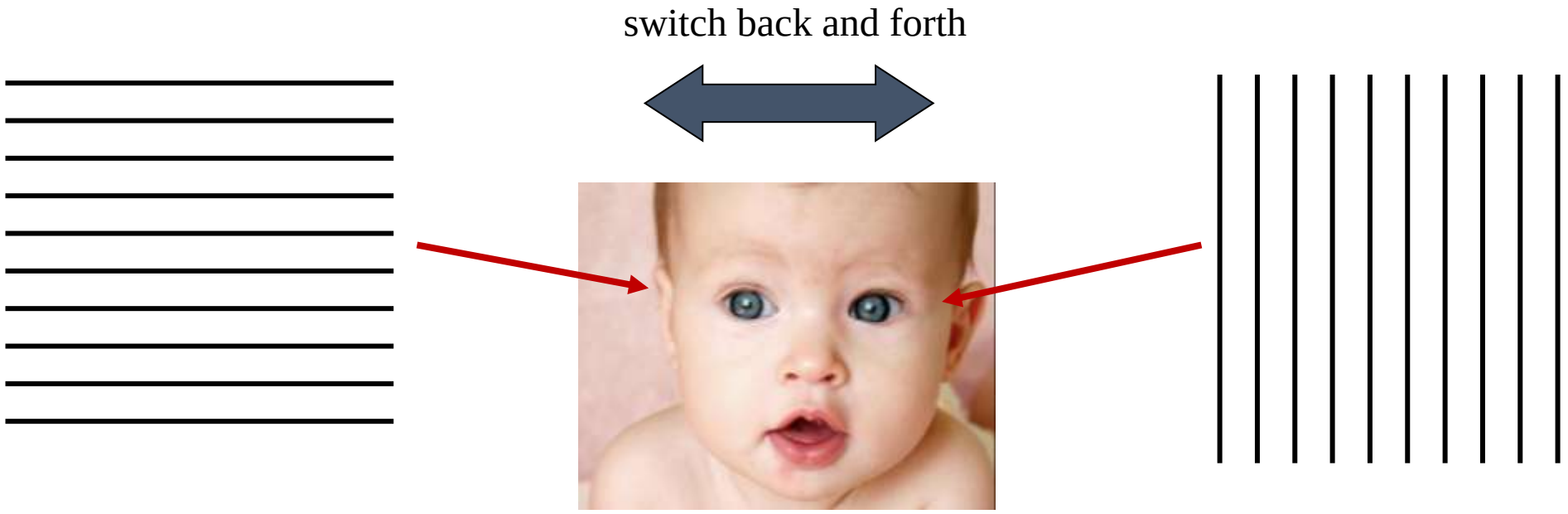
Edelman

Crick and Koch

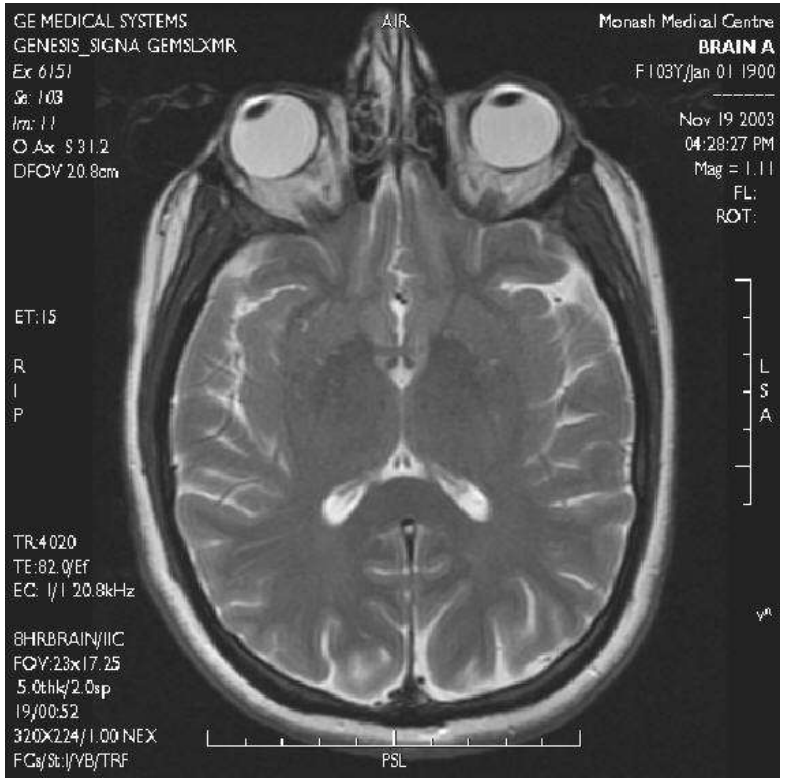
(*Wikimedia Commons*)

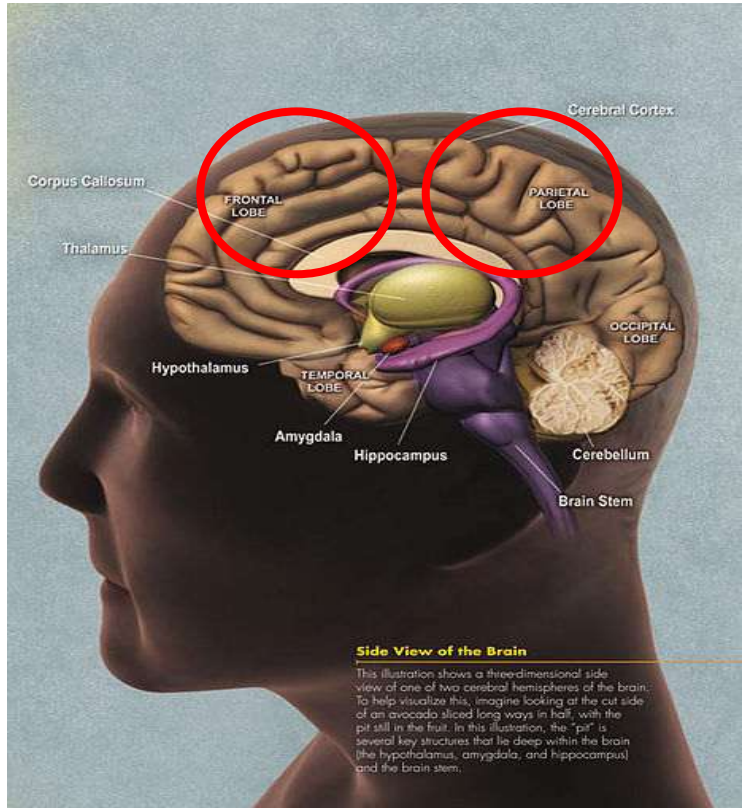
Binocular Rivalry (双目竞争) Experiment

whether the claustrum is switched on when unconscious perception of a given stimulus by a person's sensory organs turns into a conscious percept.



27 Using MRI, Eric Lumer and his colleagues at University College, London have identified the frontal and parietal areas of the cortex as the regions of the brain that become active when a person's conscious attention switches from one image to another. These two regions have a special role in focusing conscious attention





Brain imaging

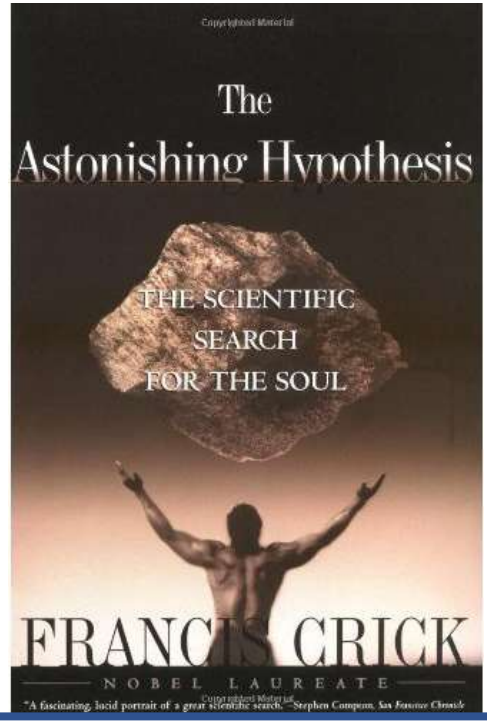
Crick's "*The Astonishing Hypothesis*"



1916-2004

"You," your joys and your sorrows, your memories and your ambitions, your sense of personal identity and free will, are in fact no more than the behavior of a vast assembly of nerve cells and their associated molecules.

"You're nothing but a pack of neurons."



Francis Crick, 1994.

Crick's contribution to biology

---put him in a class with Copernicus, Newton, Darwin, and Einstein



1916-2004

- Discovery of the double helical structure of DNA (1953)
- The nature of genetic code
- The discovery of mRNA
- Translation of mRNA into protein
- Turn to study the biological nature of consciousness (1976)
- Legitimizing of the biology of consciousness
- "Yet his intense, lifelong focus on science, on the life of mind, is something he shares with many in the scientific community, and that obsession is symbolic of science at its best!"

Hard Problem: Subjectivity

- We experience our own ideas, moods, and sensations directly.
- We can only appreciate another person’s experience indirectly by observing or hearing about it.

Hard Problem: Subjectivity

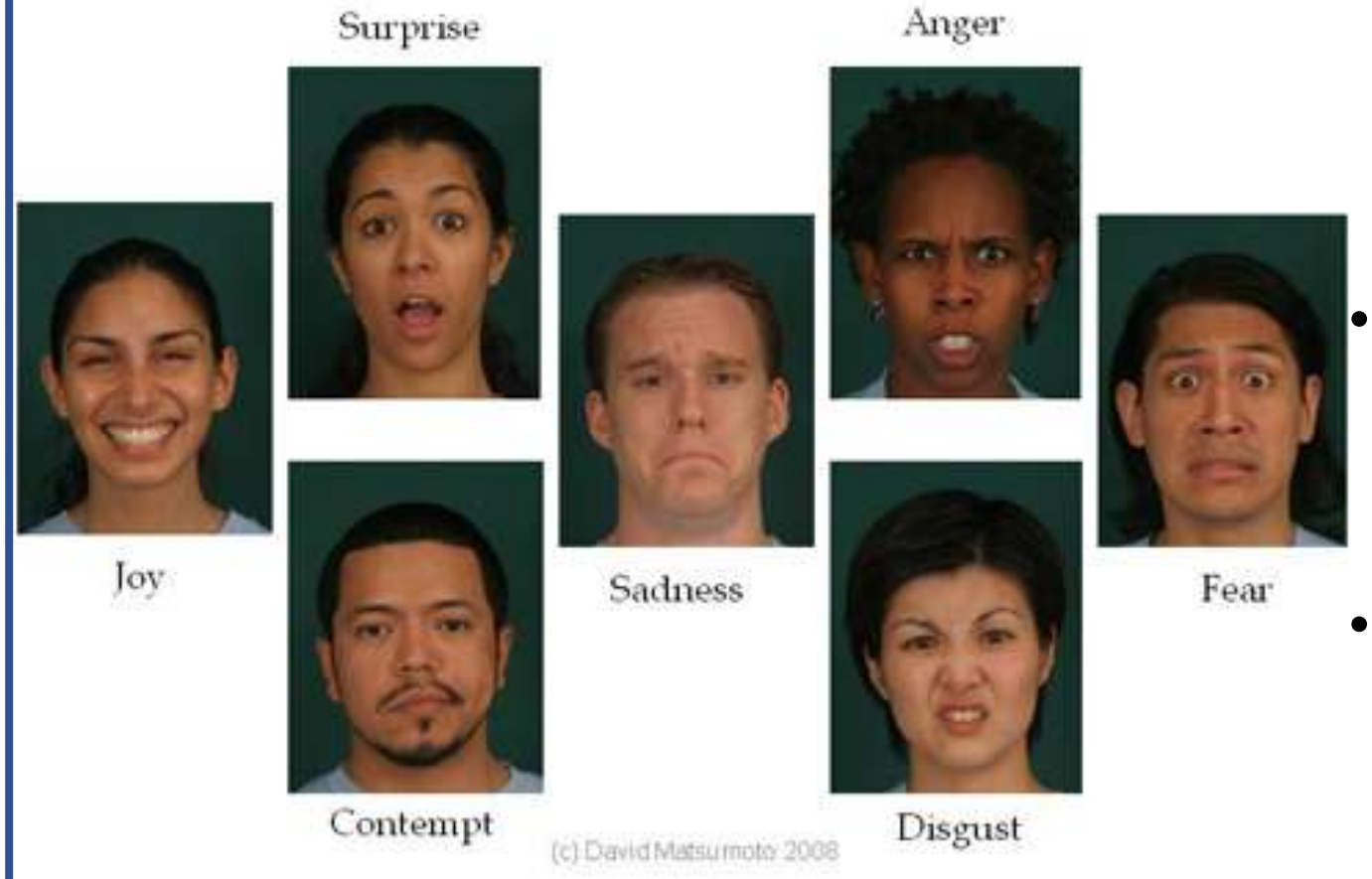
- Is your response to durian identical to my response to it?
- How does electrical activity in neurons give rise to the subjective feelings?
- We do not understand!



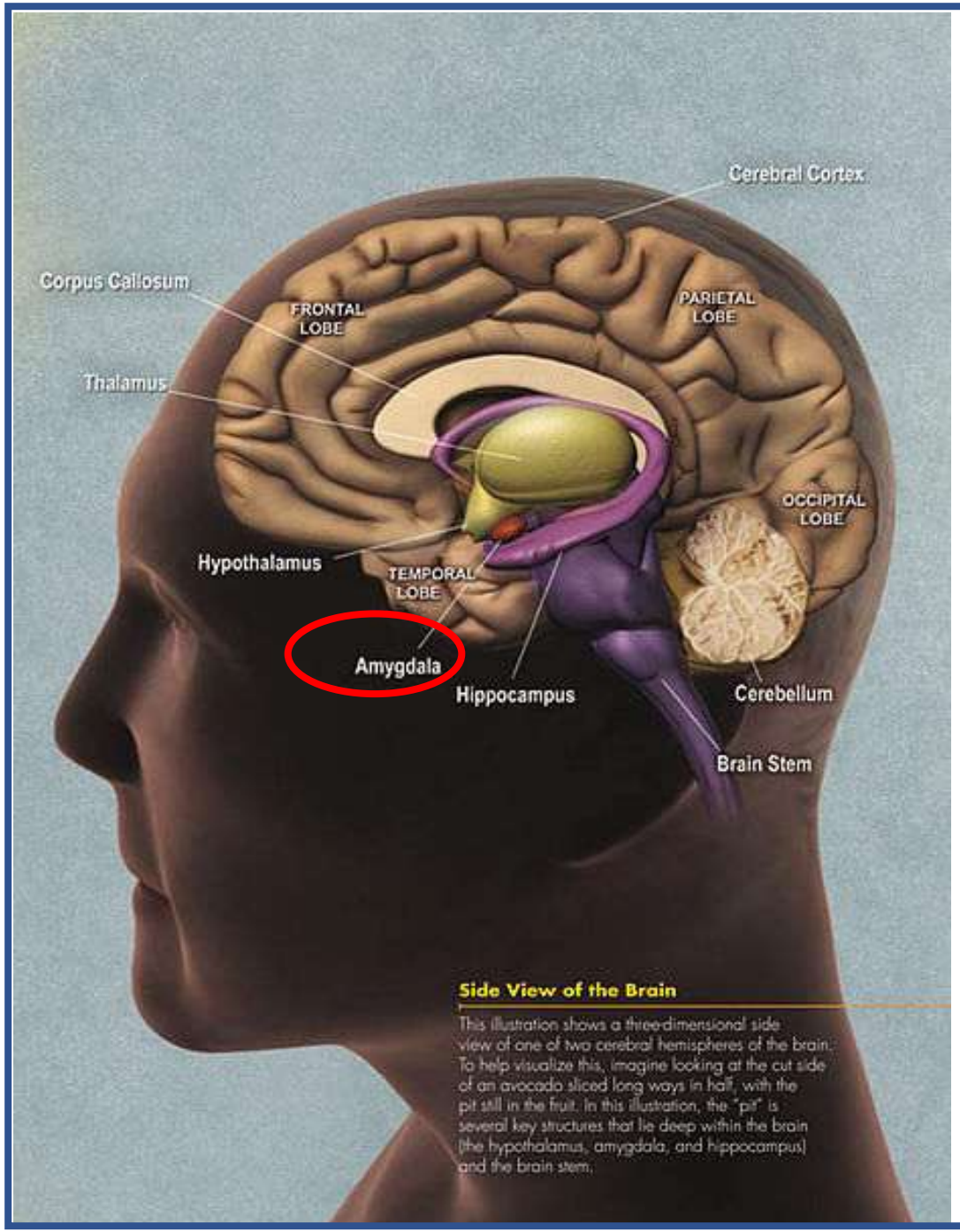
Koch's experiment



Kandel's experiment



- Irrespective of sex or culture, conscious perceptions of these seven facial expressions have virtually the same meaning to everyone.
- Volunteers perceived the fearful faces consciously (presenting for a long period) and unconsciously (presenting rapidly).
- Prominent activity was found in the amygdala.

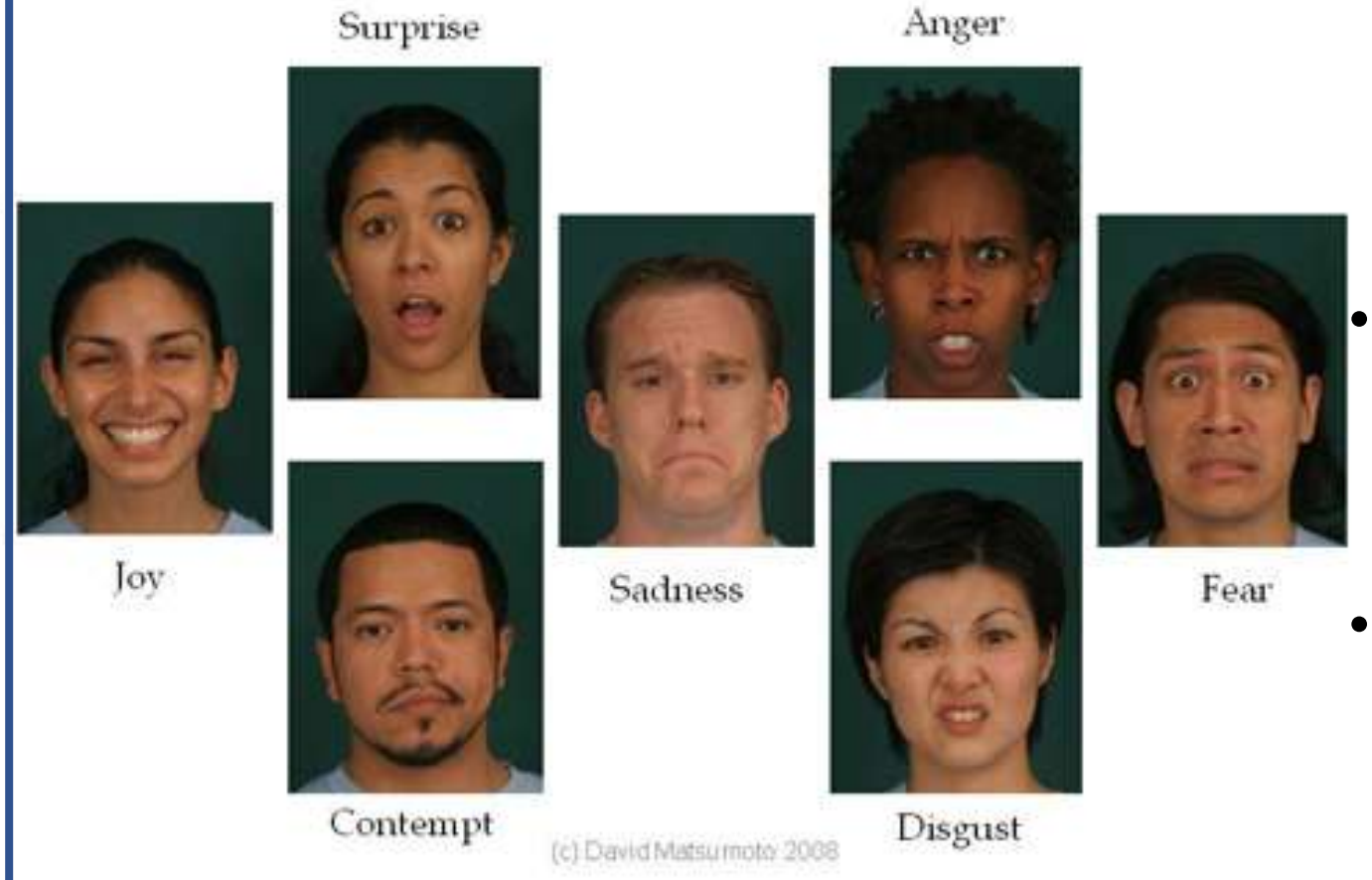


Amygdala: 杏仁体

- The amygdala represents a core fear system in the human body, which is involved in the expression of conditioned fear.
- Discovery:
  - Early in 1888, rhesus monkeys with a lesioned temporal cortex (including the amygdala) were observed to have significant social and emotional deficits.
  - Behavioral disorder: overreaction to all objects, hypo-emotionality, loss of fear…
  - Later research proved this behavioral disorder was specifically due to amygdala lesions.
- Amygdala dysfunction: anxiety, autism(自闭症), depression, posttraumatic stress disorder (PTSD), bipolar, and phobias(恐慌症).

(*Wikimedia Commons*)

Kandel's experiment



- Irrespective of sex or culture, conscious perceptions of these seven facial expressions have virtually the same meaning to everyone.

- Volunteers perceived the fearful faces consciously (presenting for a long period) and unconsciously (presenting rapidly).

- Prominent activity was found in the amygdala, conscious and unconscious stimuli affected different regions of the amygdala, they did so to differing degrees in different people, depending on their baseline anxiety.

3. Free will?

Free Will?

- Issues that central to psychoanalysis and consciousness

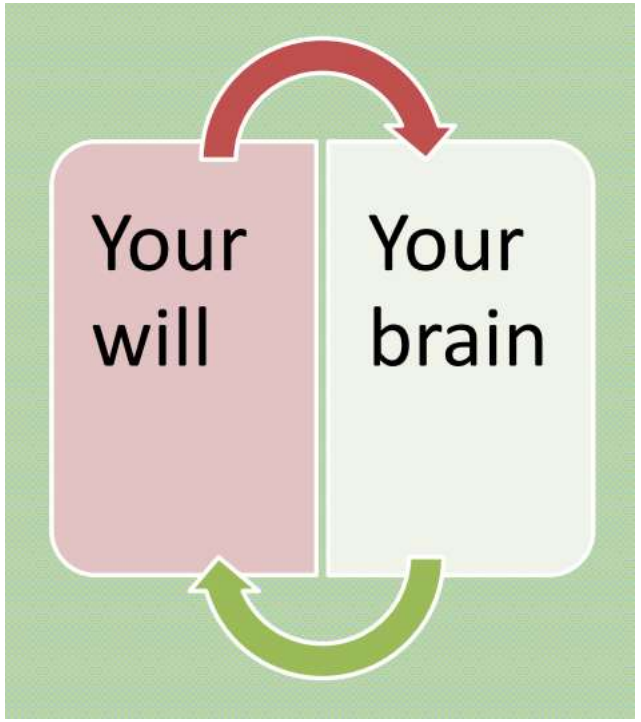
  -Free will

    -The ability to choose for ourselves what to do.

Free Will

Which one makes the decision?

Your will or your brain?



Libet's Experiment (1983)

BBC RADIO 4

FREEDOM

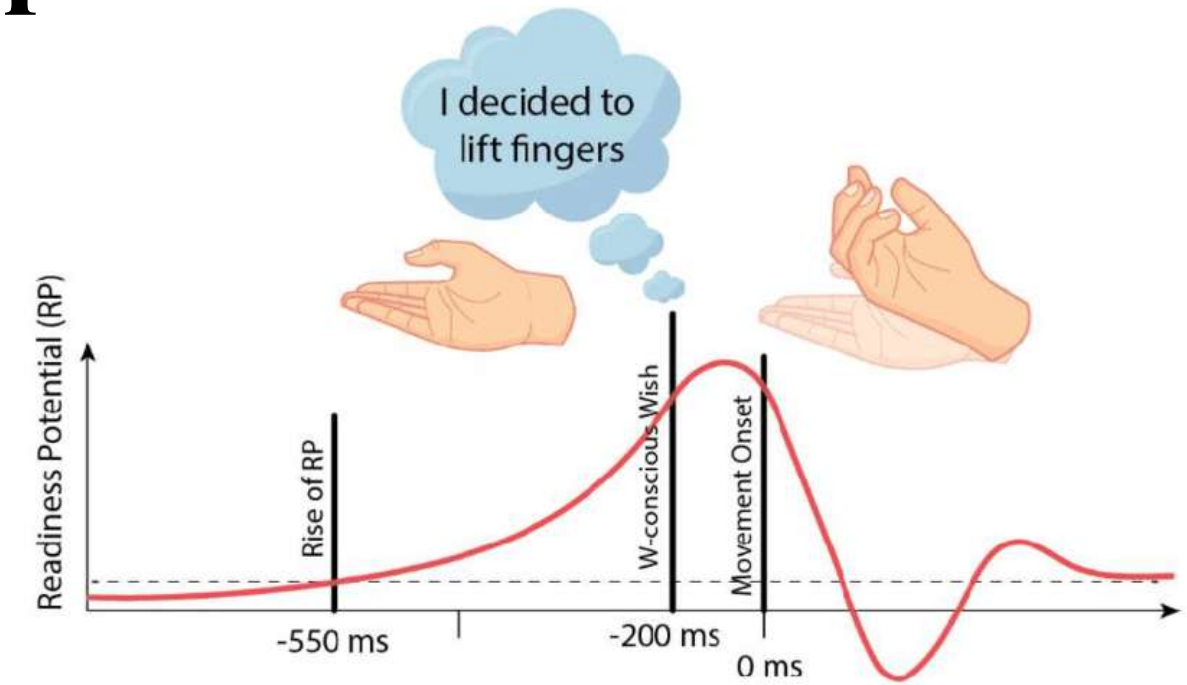
The Libets Experiments & Free Will

Narrated by  Harry Shearer

The Open University

Libet's Experiment





Libet's experiment.

- lift a finger whenever they felt the urge to do so.
- Measure the time and the electrical potentials
- Compare the time that the person will the movement with the time of readiness potential.

- Electrical potential in the brain BEFORE the volunteer's decision.

- His brain appears to prepare for movement long before he consciously decided to move.
- Does he have free will?

Reading Assignment

Read Text 8.
Quiz next tutorial (text 6,7&8).